

KELLY K. CAYLOR

Professor of Ecohydrology

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EDUCATION

2003, Ph.D., Environmental Sciences, University of Virginia
1996, B.A. *with High Distinction*, Environmental Sciences, University of Virginia

PROFESSIONAL RECORD

2025 - Present Associate Vice Chancellor for Innovation, Office of Research, UC Santa Barbara
2016 - Present Full Professor, Bren School of Environmental Science & Management, UC Santa Barbara
2016 - Present Full Professor, Dept. of Geography, University of California, Santa Barbara
2024 - 2025 Associate Vice Chancellor, Research Computing & Data, Office of Research, UC Santa Barbara
2022 - 2024 Associate Vice Chancellor, Earth, Environment, & Sustainability, Office of Research, UC Santa Barbara
2016 - 2022 Director, Earth Research Institute, University of California, Santa Barbara
2014 - 2016 Director, Environmental Studies Program, Princeton University
2013 - 2016 Associate Professor, Dept. of Civil and Environmental Engineering, Princeton University
2013 - 2016 Affiliated Faculty, Dept. of Ecology and Evolutionary Biology, Princeton University
2007 - 2012 Assistant Professor, Dept. of Civil and Environmental Engineering, Princeton University
2007 - Present Adjunct Faculty, Department of Geography, Indiana University
2005 - 2007 Assistant Professor, Department of Geography, Indiana University
2004 - 2005 Postdoctoral Research Associate, National Center for Earth Dynamics, Department of Civil and Environmental Engineering, Princeton University
2003 - 2004 Postdoctoral Research Associate, Princeton Environmental Institute, Department of Civil and Environmental Engineering, Princeton University

RESEARCH INTERESTS

Ecohydrological dynamics, Conservation ecohydrology, Coupled natural-human systems

AWARDS & HONORS

2026 Outstanding Graduate Mentor Award, University of California, Santa Barbara
2025 Fellow, American Association for the Advancement of Science
2023 Editor's Citation for Excellence in Refereeing, Water Resources Research
2018 AI for Earth Explorer, National Geographic Society
2015 Commendation List for Outstanding Teaching, School of Engineering and Applied Sciences, Princeton University
2013 E. Lawrence Keyes, Jr./Emerson Electric Co. Faculty Advancement Award, Princeton University
2012 Commendation List for Outstanding Teaching, School of Engineering and Applied Sciences, Princeton University
2011 Editor's Citation for Excellence in Refereeing, Reviews of Geophysics
2009 CAREER Award, National Science Foundation
2009 Early Career Hydrological Sciences Award, American Geophysical Union
2004 Postdoctoral Synthesis Fellowship, National Center for Earth Dynamics

AWARDS & HONORS – continued

- 2003 Clayton Postdoctoral Fellowship, Princeton Environmental Institute
2002 DuPont Fellowship, University of Virginia
2001 Earth System Science Graduate Fellowship, NASA
1998 Governor's Fellowship, University of Virginia
1996 Distinguished Major, Department of Environmental Sciences, University of Virginia

TEACHING & COURSE INSTRUCTION

- 2025 G288KC: *"Applications of Geospatial Foundation Models"*, University of California, Santa Barbara
- 2023 - EDS217: *"Python for Environmental Data Science"*, University of California, Santa Barbara
- 2019 - G136: *"Water, Energy, and Ecosystems"*, University of California, Santa Barbara
- 2018 - ESM203: *"Introduction to Earth System Science"*, University of California, Santa Barbara
- 2023 Bren MESM Project *"Collaborative Conservation Planning for the Gaviota Region"*, University of California, Santa Barbara
- 2022 Bren MESM Project *"Estimating Habitat Connectivity to Guide Wildlife Conservation and Management Planning at The Nature Conservancy's Jack and Laura Dangermond Preserve"*, University of California, Santa Barbara
- 2021 Bren MESM Project *"Estimating Habitat Connectivity to Guide Wildlife Conservation and Management Planning at The Nature Conservancy's Jack and Laura Dangermond Preserve"*, University of California, Santa Barbara
- 2020 Bren MESM Project *"Wild Pig Management at the Jack and Laura Dangermond Preserve"*, University of California, Santa Barbara
- 2019 Bren MESM Project: *"Green infrastructure for coral reef protection in Hawaii"*, University of California, Santa Barbara
- 2018 Bren MESM Project: *"Determining conservation research opportunities at the TNC Point Conception Dangermond Preserve"*, University of California, Santa Barbara
- 2017 Bren MESM Project: *"Exploring the Return on Investment Case for Natural Infrastructure Investment in the Upper Mississippi River Basin"*, University of California, Santa Barbara
- 2017 G167/ES167: *"Biogeography"*, University of California, Santa Barbara
- 2016 Freshman Seminar: *"Science, Society, and Dinner"*, with Chef Craig Shelton, Princeton University
- 2015 Summer Course on *"Isotope Hydrology Training"*, China Agricultural University
- 2015 Summer Course on *"Using UAVs for Environmental Monitoring"*, U. of Basilicata
- 2014 Summer Course on *"Modeling Spatial/Temporal Heterogeneity in Ecohydrological Systems"*, U. of Basilicata
- 2015 - 2016 EEB307/CEE307/ENV307, *"Water, Energy, and Ecosystems"*, Mpala Research Center, Princeton University
- 2013 CEE474/ENV474, *"Design and Implementation of Environmental Sensor Networks"*
- 2011 CEE474/ENV474, *"Design and Construction of Environmental Sensors"*, Princeton University
- 2009 - 2014 CEE 307, *"Field Ecohydrology"*, Mpala Research Center, Princeton University
- 2010 - 2016 ENV 201, *"Introduction to Environmental Studies"*, with Prof. David Wilcove, ENV Program, Princeton University
- 2008-2009 CEE 599, *"Environmental Biophysics"*, Princeton University
- 2008 CEE 599, *"Research Topics in Ecohydrology"*, Princeton University
- 2007 G307, *"Biogeography"*, Indiana University

TEACHING & COURSE INSTRUCTION – continued

2006	G440, “ <i>Topics in Env. Geography: Ecological Systems and Simulation</i> ”, Indiana University
2006	G440, “ <i>Topics in Env. Geography: Ecohydrology</i> ”, Indiana University
2006	G405/505, “ <i>Hydroclimatology</i> ”, Indiana University
2005 to 2007	G107, “ <i>Physical Systems of the Environment</i> ”, Indiana University

SELECTED INVITED SEMINARS & PRESENTATIONS

2024	Invited Seminar	Multi-scale measures of plant water use, Department of Ecology & Evolution, University of Connecticut, Virtual
2023	Keynote Speaker	Identifying and mitigating risk during environmental science field campaigns, Campus Safety, Health, and Environmental Management Association (CSHEMA) Annual Symposium, Santa Barbara, CA
2022	Speaker	Impacts of increasing climate variability on vegetation in dryland ecosystems, Bren School of Environmental Science and Management, Santa Barbara, CA
2022	Invited Speaker	The interacting impacts of shifting rainfall climatology and crop choice in smallholder production systems, Department of Civil and Environmental Engineering, University of Florida
2021	Keynote Speaker	A mile wide and a pixel deep: Integrating machine learning, computer vision, and satellite imagery for coupled-natural human system modeling, Helmholtz Institute conference on AI and ML in Earth & Environmental Science, Virtual
2019	Invited Speaker	Linkages between water, vegetation, and livelihoods in Sub-Saharan drylands, American Geophysical Union, San Francisco, CA
2019	Invited Speaker	Coupled natural-human dynamics in smallholder rain-fed agricultural systems, Department of Civil and Environmental Engineering, University of Florida
2019	Invited Speaker	The effects of climate variability on the structure and function of tropical vegetation, 44th New Phytologist Symposium, Accra, Ghana
2019	Keynote Speaker	The Jack and Laura Dangermond Preserve: A Perfect Place for Advancing Conservation Science, Board of Trustees Meeting, University of California, Santa Barbara
2018	Instructor	Coupled Natural-Human Systems Short Course, Ridge to Reef NRT Graduate Training Program, University of California, Irvine
2018	Invited Speaker	Session on “Vegetation dynamics and ecosystem resilience under global climate change”, Ecological Society of America Annual Meeting, New Orleans, LA
2018	Invited Speaker	Workshop on Spatial Pattern Formation in Ecosystems, Center for Theoretical Science, Princeton University
2017	Invited Speaker	Department Seminar, Plant Sciences Department, University of California, Riverside
2017	Invited Speaker	Session on “Hydrologic connectivity: linking land use changes and management to movement and transformations of resources within catchments”, ALSO Aquatic Sciences Meeting, Honolulu, HI
2016	Invited Speaker	Water for Food Global Conference, Water for Food Institute, University of Nebraska

SELECTED INVITED SEMINARS & PRESENTATIONS – continued

2015	Invited Speaker	Session on “Estimation of Spatial and Temporal Variability of Land Surface Hydro- logical and Thermal Processes”, AGU Fall Meeting, San Francisco, CA
2015	Invited Speaker	Session on “African Environments Across Space and Through Time: Integration Modern and Ancient Data for Insights into Terrestrial Ecosystem Dynamics”, GSA Annual Meeting, Baltimore, MD
2015	Invited Speaker	Department Seminar, Program in Ecology, Evolutionary Biology, and Behavior, Michigan State University
2015	Invited Speaker	SESYNC Seminar Series, National Socio-Environmental Synthesis Center (SESYNC), Annapolis, MD
2015	Invited Speaker	Princeton and the Environment, Class of '68 Symposium, Princeton, NJ
2015	Invited Lecturer	Ecohydrological applications of UAVs, Summer Course on “Using UAVs for Environmental Monitoring”, University of Basilicata, Matera, Italy
2015	Invited Participant	Conference on Water & Food Security Under Changing Environments, Center for Agricultural and Water Research in China, China Agricultural University
2015	Lead Instructor	Isotope Hydrology Training Course, Center for Agricultural and Water Research in China, China Agricultural University
2015	Invited Speaker	Seminar Series, Chinese Academy of Agricultural Sciences, Beijing, China
2015	Invited Speaker	Research Seminar Series, Earth Research Institute, University of California, Santa Barbara, CA
2015	Invited Speaker	Department Seminar, Department of Civil and Environmental Engineering, University of Michigan, Ann Arbor, MI
2015	Invited Speaker	Symposium on “Advances in Earth Observation: Enabling New Insights into Global Environmental Change”, AAAS Annual Meeting, San Jose, CA
2014	Invited Speaker	Session on “Coupled Natural Human System Research Across Scales: From Emu- lators of Global Climate Model Output to Local-Scale Processes and Feedbacks”, AGU Fall Meeting, San Francisco, CA
2014	Invited Lecturer	Ecohydrological modeling, Summer Course on “Modeling Spatial/Temporal Heterogeneity in Ecohydrological Systems”, U. of Basilicata, Matera, Italy
2014	Keynote Speaker	Annual Meeting, WISENet IGERT , Duke University, Durham, NC
2014	Invited Speaker	Workshop on “Improving Water Use Efficiency in Agriculture”, China Agricultural University, Beijing, China
2014	Invited Speaker	Arid Zone Hydrology Symposium, Texas A&M University
2013	Invited Speaker	Symposium on Hydropedology, Soil Science Society of America Meeting, Tampa, FL
2013	Invited Participant	2nd International Workshop on “Methods and Technologies for Environmental Monitoring and Modelling of Hydrologic and Hydraulic Risk”, Matera, Italy
2012	Invited Speaker	Session on “Dryland Ecohydrology: Critical Issues and Technical Advances”, AGU Fall Meeting, San Francisco, CA
2012	Invited Speaker	Department Seminar, Department of Environmental Science, Policy, and Management, University of California - Berkeley
2012	Invited Speaker	Session on “Catchment Similarity and Organisation”, EGU Annual Meeting, Vienna, Austria
2012	Invited Lecture	Stuart Country Day School, Princeton, NJ
2012	Invited Speaker	Science, Technology, Environment, and Policy Program, Woodrow Wilson School, Princeton University

SELECTED INVITED SEMINARS & PRESENTATIONS – continued

2011	Invited Speaker	Annual Board Meeting, Earthwatch, Princeton, NJ
2011	Invited Speaker	Department Seminar, Department of Civil and Environmental Engineering, University of Illinois, Urbana-Champaign
2011	Invited Speaker	Ecohydrology and Climate Change, EcoHCC'11 International Conference, Tonar, Portugal
2011	Invited Lecturer	Ecohydrological modeling, 1st Summer School on Monitoring and Modeling Surface Hydrology, Marsico Nuovo, Italy
2011	Invited Panelist	Water: Global Challenge and Local Solutions in Developing Countries, New Jersey Symphony Orchestra Winter Concert, Princeton, NJ
2010	Invited Speaker	Department Seminar, Department of Geological, Environmental, and Marine Sciences, Rider University
2009	Invited Speaker	Session on “Landscapes in Semi-Arid and Arid Environments: A Hydrological and Ecological Perspective”, AGU Fall Meeting, San Francisco, CA
2009	Invited Speaker	Seminar Series, Environmental Fluid Mechanics/Hydrology, MIT
2009	Invited Speaker	New frontiers in Hydrology, IV CNR-Princeton Workshop, Rome, Italy
2009	Invited Seminar	Seminar Series, Department of Plant Sciences, Kenyatta University of Agriculture & Technology
2008	Invited Presenter	Session on “Ecosystem resilience to changing climate patterns: the role of hydrology”, AGU Fall Meeting, San Francisco, CA
2008	Invited Presenter	Symposium, “Seeing into the soil: Noninvasive characterization of biophysical processes in the soil critical zone”, ASA-CSSA-SSSA 2008 Joint Annual Meeting, Houston, TX
2008	Invited Participant	International Ecohydrology/Ecophysiology Workshop, University of Western Australia, Perth, Australia
2008	Invited Presenter	Organized Oral Session, “Towards a synthetic theory of tree-grass coexistence in savannas”, 93rd Annual Ecological Society of America Meeting, Milwaukee, WI
2007	Invited Seminar	Seminar Series, Department of Civil and Environmental Engineering, Princeton University
2007	Invited Speaker	Monroe County Drainage Board, Bloomington, IN
2006	Invited Speaker	Session on “Bridging Hydrology, Soil Science and Ecology”, AGU Fall Meeting, San Francisco, CA
2006	Invited Speaker	Session on “Climate-soil-vegetation dynamics and their impacts on water balance and hydrological extremes”, European Geophysical Union Meeting, Vienna, Austria
2006	Invited Seminar	School of Public and Environmental Affairs, Indiana University
2005	Invited Speaker	Session on “Integrated Approaches in Hydrological Process Studies”, AGU Spring Meeting, New Orleans, LA
2005	Invited Seminar	Department of Environmental Sciences, Rutgers University
2005	Invited Seminar	Colloquia Series, Department of Geography, Indiana University
2004	Invited Seminar	Woods Hole Research Center, Woods Hole, MA
2004	Invited Seminar	Forest Watershed Seminar Series, Department of Environmental Science, Policy, and Management, University of California - Berkeley
2003	Invited Participant	Savanna Complexity: Different perceptions, Common Ground, Luiperdskloof Game Lodge, South Africa
2002	Invited Presenter	Session on Ecohydrology, AGU Spring Meeting, Washington D.C.

SELECTED SERVICE & PROFESSIONAL ACTIVITIES

2025 - Present	Interim Chair, PhD Committee, Bren School of Environmental Science and Management
2025 - Present	Associate Vice Chancellor for Innovation, Office of Research
2024 - Present	Faculty Lead, I-Corps Western Hub
2024 - Present	Editor in Chief, Earth's Future, American Geophysical Union
2026	External reviewer: Department of Environmental Sciences, Emory University
2025	Panelist, Webinar, "Reframing Research Language for Impactful and Relevant Broader Impact Statements", Consortium of Universities for the Advancement of Hydrologic Science, Inc
2025 - 2026	Moderator, Cultivating Change: Regenerative Agriculture for a Climate-Resilient Future, Mantell Symposium in Environmental Justice and Conservation Innovation
2025	Moderator, Mantell Environmental Justice Symposium, Bren School, UC Santa Barbara
2025	Tenure reviewer: University of Nevada, Reno; University of Maryland
2025	Promotion reviewer: Duke University
2025	Reviewer, Austrian Science Fund
2024	Reviewer, UC Alianza MX: Strategic Research Program
2024 - 2025	Interim Chair, MESM Curriculum Committee, Bren School of Environmental Science and Management
2024	Reviewer, NSF Division of Environmental Biology
2024	Panelist, Human-Environment and Geographical Sciences Program, National Science Foundation
2024 - 2025	Associate Vice Chancellor for Research Computing and Data, Office of Research
2024	Promotion reviewer: Clark University; Peking University
2023	Promotion reviewer: Indiana University
2023	Tenure reviewer: Stanford University
2023 - 2025	Chair, Devereux Environmental Crossroads Building Committee
2022 - 2023	Co-Convenor, AGU Fall Meeting, Session on "Conservation Ecohydrology"
2022 - 2023	Chair, Career Development and Alumni Relations Committee, Department of Geography
2022 - 2024	Associate Vice Chancellor, Office of Research
2021	Tenure reviewer: Duke University; Michigan State University
2021 - 2022	Chair, Dangermond Endowed Chair Search Committee, Office of Research
2020 - 2023	Editor, Earth's Future, American Geophysical Union
2020 - 2022	Chair, NCEAS Campus Advisory Committee
2020	Promotion reviewer: Texas A&M University
2018	Tenure reviewer: University of Toronto
2018	Co-Convenor, AGU Fall Meeting, Session on "Indicators of Plant Water Availability and Stress in Drought-Prone Forests at a Range of Spatial and Temporal Scales"
2018	Promotion reviewer: University of Washington; SUNY-ESF; University of Texas, Austin
2018 - 2021	Chair, Space Committee, Department of Geography
2017	Co-Convenor, International Association of Hydrological Sciences Conference, Session on "Innovative ICT Tools for Water Management and Science"
2017	Co-Convenor, AGU Fall Meeting, Session on "Plant-Soil Interactions Rhizosphere: Experimental and Computational Advances"
2017	Co-Convenor, AGU Fall Meeting, Session on "Advances in Ecohydrology of Water-Stressed Environments"
2017 - 2018	Program Director, Earth Research Institute Graduate Fellowship Program

SELECTED SERVICE & PROFESSIONAL ACTIVITIES – continued

2017	Tenure reviewer: University of Cape Town; University of Alabama; Indiana University
2017 - 2018	Chair, Environmental Data Science & Informatics Faculty Search Committee, Earth Research Institute
2016 - 2022	Director, Earth Research Institute
2016 - 2021	Chair, Earth Research Institute, Personnel Committee
2015 - 2016	Chair, Graduate and Postdoctoral Experiences Strategic Planning Subcommittee, Princeton University
2014	Co-Convenor, AGU Fall Meeting, Session on “The Influence of Changes in Farming Practices, Vegetation, and Land Use on Climate Adaptation, Mitigation, and Ecosystem and Socioeconomic Services”
2014 - 2016	Board Member, Princeton in Africa, Princeton NJ
2014 - 2016	Director, Graduate Studies, Department of Civil and Environmental Engineering, Princeton University
2014 - 2016	Director, Environmental Studies Program, Princeton University
2013	Co-Convenor, AGU Fall Meeting, Session on “Beyond changes in mean climate: the impacts of climate variabilities on terrestrial ecosystems”
2013	Co-Convenor, AGU Fall Meeting, Session on “Postmortem: 2012 Drought: Terrestrial Ecosystems”
2013	Participant, 2nd Annual Neureiter Science Diplomacy Roundtable, Center for Science Diplomacy, AAAS
2012	Co-Convenor, AGU Fall Meeting, Session on “Landscape System Response Under Change”
2012 - 2013	Director, Graduate Studies, Civil and Environmental Engineering, Princeton University
2012	Organizer, Workshop on “Predictions Under Change: Vision for understanding and managing water, earth, and biota in the Anthropocene”, Boulder, CO, July 2012
2011 - 2013	Associate Editor, Vadose Zone Journal
2011	Co-Convenor, Fall Meeting, Session on “Stable Isotope Fluxes in the Carbon and Water Cycles of Terrestrial Ecosystems”
2011	Co-Convenor, AGU Fall Meeting, Session on “Predictive Understanding of Coupled Interactions Among Water, Life, and Landforms”
2011	Co-Convenor, AGU Fall Meeting, Session on “Landscape System Response Under Change”
2011	Co-Convenor, AGU Fall Meeting, Session on “Laser-Based Isotope Techniques in Biogeosciences”
2011	Co-Convenor, AGU Fall Meeting, Session on “Water and Energy Footprints in the Anthropocene”
2010 - 2012	Chair, Ecohydrological Technical Subcommittee, Hydrology Section, AGU
2009 - 2012	Associate Editor, Journal of Geophysical Research - Biogeosciences
2008	Co-Convenor, AGU Fall Meeting, Session on “Hydrological Controls on Ecosystem Function”
2008 - 2009	Moderator, Colloquium on Public and International Affairs, “Eco-Friendly Development”
2007	Co-Convenor, AGU Spring Meeting, Session on “Dynamics and interaction of belowground carbon pools in dryland ecosystems”
2005	Co-Convenor, AGU Fall Meeting, Session on “Temporal and spatial variability in coupled ecological-hydrological processes”
2005	Co-Convenor, AGU Fall Meeting, Session on “Soil-water-nutrient interactions within savanna vegetation”
2004	Co-Chair, NASA ESSN Symposium, Water and Energy Session

INNOVATION & ENTREPRENEURSHIP

2025 -	Associate Vice Chancellor for Innovation	Office of Research, University of California, Santa Barbara
2024 -	PI,	NSF I-Corps Program, University of California, Santa Barbara
2023 - 2025	Campus Lead,	Climate Action Innovation Funds, University of California, Santa Barbara
2022 - 2023	Co-PI,	NSF Convergence Accelerator Track J: Predicting the effect of climate extremes on the food system to improve resilience of global and local food security
2018 - 2021	Advisor,	Princeton Climate Analytics, Princeton NJ
2015	Participant,	NSF I-Corps Entrepreneurship Program, NYC Cohort
2015	Investment,	Princeton Keller Center, Princeton Entrepreneurship Fund
2015	Award,	Princeton Keller Center, Innovation Forum
2015	Co-Founder,	Arable Labs, Inc. (www.arable.com)
2015-2017	Board Member,	Arable Labs, Inc. (www.arable.com)
2014	Co-Founder,	Pulsepod Designs, Inc.
2013	Seed Funds,	Princeton, Project X Fund

FUNDED RESEARCH

Total: \$27,758,148

2026-2028	State of California	Space Vandenberg, PI (\$1,109,156)
2025-2027	NSF	CC* Storage-Campus: Establishment of a UC Santa Barbara Data Storage Harnessing Open Research Environments (DataSHORE), PI (\$699,954)
2024-2025	National Science Foundation	NSF I-Corps Hub: West Region NPI Supplement, PI (\$133,500)
2024-2026	National Science Foundation	LSAMP BD: UC Santa Barbara California Alliance for Minority Participation, Co-PI (\$1,075,000)
2023-2025	Department of Energy	Improving ESS approaches to evapotranspiration partitioning through data fusion, Co-PI (\$250,000)
2023-2025	UC Office of the President	Broadening Participation and Seeding Climate Innovation for the Central Coast, PI (\$1,000,000)
2023-2028	Department of Defense	The climate-food-urbanization nexus and the precursors of instability in Africa, Co-PI (\$3,073,150)
2022-2024	Department of Defense	Generalizing plant functional responses to drought stress in the Southwest USA, Co-PI (\$178,284)
2022-2025	Zegar Family Foundation	Ecohydrological resilience of vegetation to water availability and drought, PI (\$672,078)
2022-2023	National Science Foundation	NSF Convergence Accelerator Track J: Predicting the effect of climate extremes on the food system to improve resilience of global and local food security, Co-PI (\$748,674)

FUNDED RESEARCH – continued

2019-2021	Zegar Family Foundation	Developing and Implementing Novel Tools for Assessing Riparian Ecosystem Resilience at the Dangermond Preserve, PI (\$321,583)
2019-2022	National Science Foundation	CNH2-L: Linkages and Interactions Between Urban Food Security and Rural Agricultural Systems, Co-PI (\$1,600,000)
2018-2018	The Nature Conservancy	Task 25: Dangermond Preserve Bren Summer Internships and Group Project, PI (\$17,000)
2018-2019	Omidyar Foundation	Developing and Scaling up the Mapping Africa Active Learning Platform, PI (\$79,838)
2018-2019	National Geographic Society	Global maps of center pivot agriculture for improving estimates of land use change and water use, PI (\$100,811)
2018-2019	National Science Foundation	WSC-Category 2 Collaborative: Impacts of Agricultural Decision Making and Adaptive Management on Food Security, PI (\$628,779)
2018-2022	Department of Defense, SERDP	Strategic Environmental Research & Development: Understanding and Assessing Riparian Habitat Vulnerability to Drought-Prone Climate Regimes on Department of Defense Bases in the Southwestern US , Co-PI (\$1,704,236)
2017-2019	Clark University	Hazards SEES: Understanding Cross-Scale Interactions of Trade and Food Policy to Improve Resilience to Drought Risk, PI (\$103,249)
2017-2020	National Science Foundation	Collaborative Research: Impacts of Dynamic Climate-Driven Water Availability on Tree Water Use and Health in Mediterranean Riparian Forests , Co-PI (\$396,566)
2017-2020	National Science Foundation	CC*Networking Infrastructure: UCSB Network Upgrade to 100 Gigabit. , PI (\$481,730)
2015-2015	National Science Foundation	I-Corps: Pulsepod: Bringing the Cloud Down to Earth, PI (\$50,000)
2015-2019	National Science Foundation	Hazards SEES: Understanding Cross-Scale Interactions of Trade and Food Policy to Improve Resilience to Drought Risk, Co-PI (\$2,519,689)
2014-2018	National Science Foundation	WSC-Category 2 Collaborative: Impacts of Agricultural Decision Making and Adaptive Management on Food Security, PI (\$1,929,565)
2013-2014	JPL	Adding precision to agricultural practice through simultaneous chlorophyll fluorescence and infrared observations, Co-PI (\$100,000)
2013-2014	JPL	Rapid forest triage by sub-canopy micro air vehicle, Co-PI (\$100,000)
2013-2015	Princeton University	Drought and the global carbon cycle, Co-PI (\$200,000)
2012-2016	National Science Foundation	Collaborative research: Quantifying the ecosystem-wide impacts of a strong interactor in African watersheds, PI (\$123,336)
2011-2014	Princeton University	A global collaborative network for coupling hydrological forecasts and food security in Sub-Saharan Africa and China , PI (\$225,000)
2011-2014	NASA	Multiscale effects of fire on long-term climate and precipitation, Co-I (\$1,149,128)

FUNDED RESEARCH – continued

2011-2016	National Science Foundation	CNH: Institutional Dynamics of Adaptation to Climate Change: Longitudinal Analysis of Snowmelt-Dependent Agricultural Systems, Co-PI (\$1,200,000)
2010-2011	Princeton University	The Importance of Green Feedbacks on Biodiversity Conservation and Management of Savanna Ecosystems, PI (\$100,000)
2010-2015	National Science Foundation	ESE: Collaborative Research: Spatial Resilience of Agriculturalists to Coupled Ecological and Hydrological Variability in Rural Zambia, PI (\$350,000)
2009-2011	National Science Foundation	US - Kenya Doctoral Dissertation Enhancement Project: The Impact of Macropores on the Spatial and Temporal Patterns of Soil Moisture in Dryland Ecosystems of Central Kenya, PI (\$15,000)
2009-2009	Li-COR Biosciences	Using the LI-6400XT in the Princeton Semester in Kenya Program: From Undergraduate Education to International Impact, PI (\$45,000)
2009-2014	National Science Foundation	CAREER: An ecohydrological framework for understanding land degradation in dryland ecosystems, PI (\$1,543,738)
2008-2013	National Science Foundation	LTREB: KLEE- scaling up and out at the Kenya Long-term Exclosure Experiment, Co-PI (\$449,800)
2007-2008	Indiana University	Quantifying organic and pharmaceutical contaminants within urbanizing streams in a central Indiana urbanizing watershed, PI (\$51,323)
2007-2008	Monroe County Drainage Board	Quantifying the impacts of development on physical and chemical hydrology of central Indiana streams, PI (\$96,000)
2007-2010	National Science Foundation	Collaborative Proposal: Distribution and Dynamics of Belowground Carbon in Savannas, PI (\$208,344)
2007-2010	National Science Foundation	HSD: Dynamics of Reforestation in Coupled Social-Ecological Systems: Modeling Land-Use Decision Making and Policy Impacts, Co-PI (\$755,115)
2006-2010	National Science Foundation	IRES: US-South Africa International Research Experience for Students – Impacts of Disturbance on Above- and Below-ground Structure of Southern African Savannas, Co-PI (\$623,522)
2005-2007	NASA	Sources, Transports, and Impacts of Southern African Aerosols: Synthetic Case Studies Using Terra & Aqua Satellite Products, Co-I (\$650,000)
2005-2007	NASA	Hydrologic & Nutrient Controls on the Structure and Function of Southern African Savannas: A Multi-Scale Approach, Co-I (\$900,000)

RESEARCH MANUSCRIPTS (Google Scholar h-index: 61, Citations: 16,648)

- ▲ indicates current or former graduate advisee
- ◆ indicates current or former postdoctoral advisee
- ‡ indicates current or former undergraduate group member
- indicates current or former graduate committee member
- * indicates current or former visiting scholar

2026

159. ▲Morgan, B., Trugman, A.T., **Caylor, K.K.** (2026) Co-regulation of water use and canopy temperature

in desert trees. *Agricultural & Forest Meteorology*. doi:10.1016/j.agrformet.2025.110929.

158. Chen, H., Good, S.P., Zahn, E., Bou-Zeid, E., **Caylor, K.K.**, Fiorella, R.P., Haagsma, M., Wang, L. (2026) Shifts in evapotranspiration components during heatwaves alter surface cooling. *Earth's Future*. doi:10.1029/2025EF006562.
157. Chen, H., Good, S.P., **Caylor, K.K.**, Fiorella, R.P., Wang, L. (2026) A hybrid Penman-Monteith and machine learning model for simulation evapotranspiration and its components. *Journal of Hydrology*. doi:10.1016/j.jhydrol.2026.134985.

2025

156. •Meerdink, S.K., Roberts, D.A., King, J.Y., Roth, K.L., Gader, P.D., **Caylor, K.K.** (2025) Using hyperspectral and thermal imagery to monitor stress of Southern California plant species during the 2013-2015 drought. *ISPRS Journal of Photogrammetry and Remote Sensing*. doi:10.1016/j.isprsjprs.2025.01.015.
155. Xenopoulos, M.A., Beal, L., Caprarelli, G., **Caylor, K.K.**, Destouni, G., Duan, Q., Hauk, S.A., , Huber, M., Karnaуска, K.B., Lajtha, K., Lugas, N., Montanari, A., Nguyen, H., Parrish, J., Rowan, A., Santos, II., Schubert, A., Wyssession, M. (2025) Commitment to advance excellence and inclusion in the Earth and space sciences scholarly publications. *AGU Advances*. doi:10.1029/2025AV001726.
154. •Araki, R., ▲Morgan, B., McMillan, H.K., **Caylor, K.K.** (2025) Nonlinear soil moisture loss function reveals vegetation responses to water availability. *Geophysical Research Letters*. doi:10.1029/2024GL111403.
153. ▲Morgan, B., •Araki, R., Trugman, A., **Caylor, K.K.** (2025) Ecological and hydroclimatic determinants of vegetation water-use strategies. *Nature Ecology & Evolution*. doi:10.1038/s41559-025-02810-8.
152. Baylis, K., Lentz, E., **Caylor, K.K.**, Gu, M., Gunderson, C., Haigh, T., Hayes, M., Lahr, H., Maxwell, D., Funk, C. (2025) Five Lessons for Closing the Last Mile: How to Make Climate Decision Support Actionable. *Earth's Future*. doi:10.1029/2024EF005799.
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2024

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164. Zahn, E., Bou-Zeid, E., Good, S.P., Katul, G.G., Thomas, C.K., Ghannam, K., Smith, J.A., Chamecki, M., Dias, N.L., Fuentes, J.D., Alfieri, J.G., Kwon, H., **Caylor, K.K.**, Gao, Z., Soderberg, K., Bambach, N.E., Hipps, L., Prueger, J.H., Kustas, W.P. (2021) Experimental and Numerical Investigation of Flux Partitioning methods for water vapor and carbon dioxide. American Geophysical Union Fall Meeting, New Orleans, LA.
163. McMahon, C., Roberts, D.A., Singer, M.B., **Caylor, K.K.**, Stella, J.C. (2021) Phenological Classification and Atmospheric Drought Response of Riparian Vegetation in Drylands of the Southwestern United States. American Geophysical Union Fall Meeting, New Orleans, LA.
162. Kibler, C.L., Roberts, D.A., Trugman, A.T., **Caylor, K.K.**, Stella, J.C., Singer, M.B. (2021) Drought Responsiveness of Evapotranspiration in Dryland Riparian Vegetation. American Geophysical Union Fall Meeting, New Orleans, LA.
161. Stella, J.C., Williams, J., Pelletier, L., Kibler, C.L., Rohde, M.M., Singer, M.B., Roberts, D.A., **Caylor, K.K.**, Lambert, A. (2021) Water Stress in Riparian Woodlands from Groundwater Decline and Climate Change—Ecosystem Indicators at Multiple Scales. American Geophysical Union Fall Meeting, New Orleans, LA.
160. Warter, M.M., Singer, M.B., Stella, J.C., **Caylor, K.K.**, Roberts, D.A., Sabathier, R. (2021) Hydroclimatic Controls on Phenology for Functional Plant Types in Drylands of the Southwest USA. American Geophysical Union Fall Meeting, New Orleans, LA.
159. Sabathier, R., Singer, M.B., Stella, J.C., **Caylor, K.K.**, Roberts, D.A., Jaeger, K.L., Olden, J.D. (2021) Flow Permanence Along Ephemeral Streams in the Huachuca Sky Island, Arizona and its Consequences for Vegetation Distribution. American Geophysical Union Fall Meeting, New Orleans, LA.
158. Morgan, B., **Caylor, K.K.** (2021) A novel UAV-based approach for estimating evapotranspiration from thermal imagery at multiple scales. American Geophysical Union Fall Meeting, New Orleans, LA.

157. Green, R.K., **Caylor, K.K.** (2021) Coupled vegetation-hydroclimatic dynamics across East Africa, drivers of prediction skill in cropland regions and implications for drought early warning.. American Geophysical Union Fall Meeting, New Orleans, LA.
156. Xiong, S., Vergopolan, N., Cecil, M., **Caylor, K.K.**, Krell, N., Konar, M., Evans, T., Baylis, K., Estes, L.D. (2021) Leveraging High-resolution Gridded Datasets and AquaCrop to Improve Remote Sensing-based Estimates of Smallholder Maize Yields. American Geophysical Union Fall Meeting, New Orleans, LA.

2020

155. Forbes, E.S., **Caylor, K.K.**, Hirsch, M.E., Schimel, J.P., Young, T., Young, H.S. (2020) Wild robots: Developing DIY technology to investigate soil carbon flux in a long-term, landscape-scale, large herbivore enclosure experiment in a central Kenya savanna. American Geophysical Union Fall Meeting, Virtual.
154. Tuholske, C., **Caylor, K.K.**, Funk, C., Verdin, A., Sweeney, S., Peterson, P., Evans, T., Grace, K. (2020) Global Urban Population Exposure to Extreme Heat. American Geophysical Union Fall Meeting, Virtual.
153. Sheffield, J., Vergopolan, N., Xiong, S., Estes, L.D., Wanders, N., Chaney, N.W., Wood, E.F., Konar, M., **Caylor, K.K.**, Beck, H., Gatti, N., Evans, T. (2020) Improving drought monitoring and crop yield prediction with field-scale soil moisture. American Geophysical Union Fall Meeting, Virtual.
152. Green, R.K., **Caylor, K.K.**, Funk, C.C., Roberts, D.A. (2020) Seasonal Vegetation-Hydrological Coupling across Land Covers in East Africa. American Geophysical Union Fall Meeting, Virtual.
151. Krell, N., Morgan, B., Gower, D., **Caylor, K.K.** (2020) Consequences of variable rainfall on farming outcomes for dryland maize farmers. American Geophysical Union Fall Meeting, Virtual.

2019

150. Estes, L.D., Song, L., Ye, S., Avery, R., McRitchie, D., Debats, S., Xiong, S., Eastman, R., Woodard, T., **Caylor, K.** (2019) Integrating humans and machines to map smallholder-dominated agricultural frontiers. 4th Open Science Meeting of the Global Land Programme, Bern, Switzerland.
149. Green, R.K., **Caylor, K.K.**, Funk, C.C., Roberts, D.A. (2019) Drought projection and modeling causal feedbacks with Earth Observations in East Africa. American Geophysical Union Fall Meeting, San Francisco.
148. Morgan, B., Mayes, M.T., **Caylor, K.K.** (2019) Improved assessment of drylands riparian vegetation water use at leaf-to-landscape scales using unmanned aerial vehicle (UAV)-based thermal remote sensing. American Geophysical Union Fall Meeting, San Francisco.
147. Vergopolan, N., Xiong, S., Sheffield, J., Wood, E.F., Evans, T.P., **Caylor, K.K.**, Estes, L.D. (2019) The spatiotemporal scales of drought and its impacts on field-scale agricultural yields. American Geophysical Union Fall Meeting, San Francisco.
146. Mayes, M.T., Morgan, B., **Caylor, K.K.**, Singer, M.B., Stella, J.C., Roberts, D.A. (2019) Monitoring dryland riparian vegetation water use and stress via ECOSTRESS. American Geophysical Union Fall Meeting, San Francisco.
145. Estes, L.D., Ye, S., Song, L., Avery, R.B., McRitchie, D., Eastman, R., Debats, S.R., **Caylor, K.K.** (2019) Improving maps of smallholder-dominated croplands through tight integration of human and machine intelligence. American Geophysical Union Fall Meeting, San Francisco.
144. Tuholske, C., Funk, C.C., **Caylor, K.K.**, Evans, T., Sweeney, S. (2019) A fine-resolution, global comparison of urban exposure to extreme temperature. American Geophysical Union Fall Meeting, San Francisco.
143. Krell, N., Davenport, F., Peterson, S., Shukla, S., Husak, G.J., Turner, W., Funk, C.C., **Caylor, K.K.** (2019) To what extent does climate variability explain farmers' planting decisions in central Kenya?. American Geophysical Union Fall Meeting, San Francisco.

142. Sabathier, R., Singer, M.B., Stella, J.C., **Caylor, K.K.**, Roberts, D.A. (2019) The greening and browning of the Arizona desert in response to climate-controlled water availability. American Geophysical Union Fall Meeting, San Francisco.
141. Forbes, E.S., **Caylor, K.K.**, Young, T., Hirsch, M., Young, H.S. (2019) Animal, vegetable, mineral: Identifying the indirect effects of large vertebrate herbivores and land use on carbon cycling in a Kenyan savanna ecosystem. American Geophysical Union Fall Meeting, San Francisco.
140. Ribeiro, F., Davis, F.W., Roberts, D.A., **Caylor, K.K.**, Hess, L.L., Daldegan, G.A., Brum, FT. (2019) Is Forest Code legislation protecting Neotropical savanna habitats in agricultural landscapes? A high-resolution Gap analysis of the Brazilian Cerrado. American Geophysical Union Fall Meeting, San Francisco.
139. Warter, M.M., Singer, M.B., Roberts, D.A., Cuthbert, M.O., **Caylor, K.K.**, Sabathier, R., Stella, J.C. (2019) Onset and propagation of the 2012-2016 drought in southern California. American Geophysical Union Fall Meeting, San Francisco.
138. **Caylor, K.K.**, Evans, T.P., Estes, L.D., Sheffield, J., Vergopolan, N., Krell, N., Gower, D., Guido, Z. (2019) Linkages between water, vegetation, and livelihoods in Sub-Saharan drylands. American Geophysical Union Fall Meeting, San Francisco.

2018

137. Tuholske, C., Andam, K., Blekking, J., **Caylor, K.**, Evans, T. (2018) Towards assessing urban food security and nutrition in West Africa. American Association of Geographers Annual Meeting, New Orleans, LA.
136. Estes, L., **Caylor, K.K.**, Debats, S., Eastman, R., Thompson, D.R. (2018) Integrating humans and machines to map smallholder-dominated agricultural frontiers. Association of American Geographers Annual Meeting, New Orleans, LA.
135. Lopus, S., Waldman, K., Schumacher, J., **Caylor, K.**, Evans, T. (2018) Population Growth and Climate Change as Simultaneous Threats to African Smallholder Farming. Population Association of America Annual Meeting, Denver, CO.
134. Manfreda, S., Ben Dor, E., **Caylor, K.K.**, Ciruolo, G., De Lima, I., De Lima, J.L., Estes, L., Frances, F., Helman, D., Kohv, M., Lucas, R., Madrigal, V.P., Mallinis, G., Maltese, A., McCabe, M., Miller, P., Perks, M., Ruiz-Perez, G., Tauro, F., Toth, B. (2018) On the use of unmanned aerial systems for environmental monitoring. European Geosciences Union, General Assembly, Vienna, Austria.
133. Mayes, M., Sylvester, A., **Caylor, K.K.** (2018) UAS imagery of debris basin volumes and infrastructure supports watershed-scale assessment and preparation for changing food and debris flow hazards in Santa Barbara county. Joint 70th Rocky Mountain Annual Section/114th Cordilleran Annual Section Meeting.
132. **Caylor, K.K.** (2018) Multi-scale dynamics of vegetation responses to changing rainfall variability in dryland regions. Ecological Society of America, New Orleans, LA.
131. Lopus, S.W., Guido, Z., **Caylor, K.**, Evans, T. (2018) Regional migration, global climate change, and the future of irrigation on Kenyan farms. International Conference on Migration, Environment and Climate: What risk inequalities?, Paris, France.
130. Tuholske, C., **Caylor, K.K.** (2018) Fusing Remote Sensing and Demography to Assess which Sub-Saharan African Cities Are Growing the Fastest. American Geophysical Union Fall Meeting, Washington, DC.
129. Avery, R., **Caylor, K.K.**, Estes, L.D., Eastman, R., Ye, S., Song, L., Zhang, K., Xiong, S., McRitchie, D., Woodard, T. (2018) A Convolutional Neural Network Approach to Segmenting Smallholder Agriculture. American Geophysical Union Fall Meeting, Washington, DC.

128. Cecil, M., Baylis, K., Blekking, J., **Caylor, K.K.**, Evans, T.P., Konar, M., Sheffield, J., Vergopolan, N., Waldman, K., Zhao, Y., Estes, L.D. (2018) Quantifying the Effect of Farmer Management Decisions on Maize Yield in Zambia. American Geophysical Union Fall Meeting, Washington, DC.
127. Estes, L.D., Debats, S., McRitchie, D., Eastman, R., Song, L., Woodard, T., Xiong, S., Ye, S., Zhang, K., Avery, R., **Caylor, K.K.** (2018) Using active learning to quantify how training data errors impact classification accuracy over smallholder-dominated agricultural systems. American Geophysical Union Fall Meeting, Washington, DC.
126. **Caylor, K.K.**, Forbes, E.S., Lewin, G., Hirsh, M. (2018) Reimagining high-resolution ecosystem monitoring with low-cost autonomous sensing. American Geophysical Union Fall Meeting, Washington, DC.
125. Mayes, M.T., **Caylor, K.K.** (2018) Seasonal and diurnal drone and ground-based thermal, multispectral and hyperspectral imaging to quantify responses of California oak woodland productivity and evapotranspiration to extreme climate conditions. American Geophysical Union Fall Meeting, Washington, DC.

2017

124. **Caylor, K.K.** (2017) Dryland feedbacks between biogeochemistry, plants, and surface hydrological dynamics. Association for the Sciences of Limnology and Oceanography, Honolulu, HI.
123. Gerlein-Safdi, C., Koochafkan, M., Chung, M., Rockwell, F.E., Thompson, S.E., **Caylor, K.K.** (2017) Effects of dew deposition on transpiration and carbon uptake in leaves. American Geophysical Union Fall Meeting.
122. Gower, D., Zeng, Z., **Caylor, K.K.**, Wood, E.F. (2017) Simulating the impact of water storage on agricultural intensification and deforestation in Northern Thailand. American Geophysical Union Fall Meeting.
121. Ribeiro, F., Roberts, D.A., Hess, L.L., Davis, F.W., **Caylor, K.K.**, Nackoney, J., Antunes Daldegan, G. (2017) High spatial resolution mapping of land cover types in a priority area for conservation in the Brazilian savanna. American Geophysical Union Fall Meeting.
120. Mayes, M.T., **Caylor, K.K.**, Ehlmann, B.L., Greenberger, R.N., Estes, L.D. (2017) Is deciduousness a key to climate resilience among iconic California savanna oak species? Relating phenological habits to seasonal indicators of tree physiological and water stress across field, hyperspectral, drone (UAS)-based multispectral and thermal image data. American Geophysical Union Fall Meeting.
119. Krell, N., Evans, T.P., Estes, L.D., **Caylor, K.K.** (2017) Real-time monitoring of smallholder farmer responses to intra-seasonal climate variability in central Kenya. American Geophysical Union Fall Meeting.

2016

118. Ruiz-Pérez, G., Manfreda, S., **Caylor, K.**, Francés, F. (2016) Automatic calibration of a parsimonious ecohydrological model in a sparse basin using the spatio-temporal variation of the NDVI. European Geosciences Union, General Assembly.
117. DeCarlo, K., **Caylor, K.** (2016) Faunal Drivers of Soil Flux Dynamics via Alterations in Crack Structure. EGU General Assembly Conference.
116. Mayes, M.T., Estes, L.D., Gago, X., Debats, S.R., **Caylor, K.K.**, Manfreda, S., Oudemans, P., Ciraolo, G., Maltese, A., Nadal, M. (2016) Using Small Drone (UAS) Imagery to Bridge the Gap Between Field- and Satellite-Based Measurements of Vegetation Structure and Change. American Geophysical Union Fall Meeting.
115. Evans, T.P., **Caylor, K.K.**, Estes, L.D., Plale, B.A., Attari, S., Waldman, K. (2016) Rapid Monitoring of Drought Impacts on Small-Scale Farms in Africa through Integration of Farmer SMS data and Environmental Sensors. American Geophysical Union Fall Meeting.

114. Estes, L.D., Debats, S.R., **Caylor, K.K.**, Evans, T.P., Gower, D., McRitchie, D., Searchinger, T., Thompson, D.R., Wood, E.F., Zeng, L. (2016) Combining Human and Machine Learning to Map Cropland in the 21st Century's Major Agricultural Frontier. American Geophysical Union Fall Meeting.
113. Evans, T.P., Estes, L.D., **Caylor, K.K.**, McCord, P.F., Gower, D., Konar, M., Baylis, K., Waldman, K., Blekking, J., Schlachter, T. (2016) Harmonizing Social and Environmental Dynamics in Earth Systems Modeling. American Geophysical Union Fall Meeting.
112. Gower, D., McCord, P.F., **Caylor, K.K.**, Dell'Angelo, J., Evans, T.P. (2016) An ecohydrologic framework for simulating catchment constraints on smallholder irrigation systems in drylands. American Geophysical Union Fall Meeting.
111. Gerlein-Safdi, C., Frolking, S.E., **Caylor, K.K.** (2016) Characterization of canopy dew formation in tropical forests using active microwave remote sensing. American Geophysical Union Fall Meeting.
110. DeCarlo, K.F., **Caylor, K.K.** (2016) Linking Carbon Flux Dynamics and Soil Structure in Dryland Soils. American Geophysical Union Fall Meeting.
109. Zeng, Z., Tian, D., Estes, L.D., Evans, T.P., Wood, E.F., **Caylor, K.K.** (2016) Can we forecast farmers' yields? The relationships between rainfall variability, farmers' expectations, and actual yields in a tropical dryland. American Geophysical Union Fall Meeting.
108. Colliander, A., Jackson, T.J., Bindlish, R., Chan, S., Das, N.N., Kim, S., Cosh, M.H., Dunbar, R.S., Asanuma, J., Aida, K., and others, including, **Caylor, K.K.** (2016) Assessment of SMAP L2/L3 Soil Moisture Products using In Situ Based Core Validation Sites. American Geophysical Union Fall Meeting.

2015

107. Debats, S.R., Estes, L.D., **Caylor, K.K.** (2015) Interactively Improving Agricultural Field Mapping in Sub-Saharan Africa with Crowd-Sourcing and Active Learning. American Geophysical Union Fall Meeting.
106. Manfreda, S., Baldwin, D.C., Keller, K., **Smithwick, E.A.H.**, Caylor, K.K. (2015) Predicting root zone soil moisture with satellite near-surface moisture data in semiarid environments. American Geophysical Union Fall Meeting.
105. Spiegel, M.P., Estes, L.D., **Caylor, K.K.**, Searchinger, T. (2015) Reconciling Agricultural Needs with Biodiversity and Carbon Conservation in a Savanna Transformation Frontier. American Geophysical Union Fall Meeting.
104. Evans, T.P., **Caylor, K.K.**, Estes, L.D., McCord, P.F., Attari, S., Sheffield, J. (2015) Modeling Local vs. Global Dimensions of Food Security in Sub-Saharan Africa. American Geophysical Union Fall Meeting.
103. DeCarlo, K.F., **Caylor, K.K.** (2015) Faunal Influences on Fracture-Induced Carbon Flux Dynamics in Dryland Soils. American Geophysical Union Fall Meeting.
102. Colliander, A., Jackson, T.J., Chan, S., Das, N.N., Kim, S., Dunbar, R.S., Bindlish, R., Dang, L.B., Berg, A.A., Rowlandson, T.L., and others, including, **Caylor, K.K.** (2015) SMAP L2/L3 soil moisture product validation using in situ based core validation sites. American Geophysical Union Fall Meeting.
101. Good, S.P., Wang, L., **Caylor, K.K.** (2015) Global optimum vegetation rain water use is determined by aridity. American Geophysical Union Fall Meeting.
100. Krell, N., DeCarlo, K.F., **Caylor, K.K.** (2015) Analysis of Biophysical Mechanisms of Gilgai Microrelief Formation in Dryland Swelling Soils Using Ultra-High Resolution Aerial Imagery. American Geophysical Union Fall Meeting.
99. **Caylor, K.K.**, Wayland, H., Scanlon, T.M. (2015) A dynamic, non-steady state approach for partitioning of soil evaporation and plant water use at landscape scales. American Geophysical Union Fall Meeting.

98. Tian, D., Estes, L.D., Evans, T.P., **Caylor, K.K.**, Sheffield, J., Wood, E.F. (2015) Impacts of intra-seasonal agricultural decision-making and forecast information on maize production in Zambia. American Geophysical Union Fall Meeting.
97. Evans, T.P., Attari, S., Plale, B.A., **Caylor, K.K.**, Estes, L.D., Sheffield, J. (2015) New Approaches to Capture High Frequency Agricultural Dynamics in Africa through Mobile Phones. American Geophysical Union Fall Meeting.
96. Gerlein-Safdi, C., Sinkler, C.J., **Caylor, K.K.** (2015) Foliar Shielding: How Non-Meteoric Water Deposition Helps Leaves Survive Drought by Reducing Incoming Energy. American Geophysical Union Fall Meeting.
95. Sheffield, J., He, X., Wanders, N., Wood, E.F., Ali, A., Olang, L., Estes, L.D., **Caylor, K.K.**, Evans, T.P. (2015) Translating the potential of hydrological forecasts into improved decision making in African regions. American Geophysical Union Fall Meeting.
94. Chang, E., **Caylor, K.K.**, Gerlein-Safdi, C., Wolf, A. (2015) Solid Phase Extraction Using C-18 Sorbents to Treat Organics in Water and Eliminate Spectral Interference in Isotope Ratio Infrared Spectroscopy. American Geophysical Union Fall Meeting.
93. Wang, L., Manzoni, S., Ravi, S., Riveros-Iregui, D.A., **Caylor, K.K.** (2015) Dynamic interactions of ecohydrological and biogeochemical processes in water-stressed environments. American Geophysical Union Fall Meeting.
92. Gower, D., **Caylor, K.K.**, McCord, P.F., Evans, T.P. (2015) Modeling the impacts of regulatory frameworks on self-organization in dryland agricultural systems. American Geophysical Union Fall Meeting.

2014

91. Pizzolla, T., Manfreda, S., **Caylor, K.**, Gioia, A., Iacobellis, V. (2014) The soil water balance in a mosaic of clumped vegetation. EGU General Assembly Conference.
90. **Caylor, K.K.**, Wolf, A., Siegfried, B. (2014) Addressing critical environmental data gaps via low-cost, real-time, cellular-based environmental monitoring. American Geophysical Union Fall Meeting.
89. Wang, L., Good, S.P., **Caylor, K.K.** (2014) Global Synthesis of Vegetation Control on Evapotranspiration Partition. American Geophysical Union Fall Meeting.
88. Debats, S.R., Luo, D., Estes, L.D., Fuchs, T., **Caylor, K.K.** (2014) Mapping Agricultural Fields in Sub-Saharan Africa with a Computer Vision Approach. American Geophysical Union Fall Meeting.
87. Wayland, H., **Caylor, K.K.** (2014) Continuous Observations of Leaf Transpiration Isotopic Composition in Two Dryland Species Highlight Sensitivity to Changes in Irradiance and Soil Water Content.. American Geophysical Union Fall Meeting.
86. DeCarlo, K.F., Spiegel, M., **Caylor, K.K.** (2014) Biological Dimensions of Crack Morphology in Dryland Soils. American Geophysical Union Fall Meeting.
85. **Caylor, K.K.**, Evans, T.P., Estes, L.D., Sheffield, J., Plale, B.A., Attari, S. (2014) Impacts of agricultural decision making and adaptive management on food security in africa. American Geophysical Union Fall Meeting.
84. Sinkler, C.J., Gerlein-Safdi, C., **Caylor, K.K.** (2014) Modeling Foliar Uptake in Colocasia Esculenta Using High Resolution Maps of Leaf Water Isotopes. American Geophysical Union Fall Meeting.
83. Estes, L.D., **Caylor, K.K.**, Chaney, N., Herrera-Estrada, J.E., Sheffield, J., Wood, E.F. (2014) Recents declines in potential evapotranspiration over South Africa: potential causes and implications for maize yield and irrigation demand. American Geophysical Union Fall Meeting.

82. Riihimäki, C.A., **Caylor, K.K.**, Wilcove, D.S. (2014) Planetary boundaries and environmental citizenship: enhancing environmental science through the Princeton University Science and Engineering Education Initiative. American Geophysical Union Fall Meeting.
81. Gerlein-Safdi, C., Sinkler, C.J., **Caylor, K.K.** (2014) Creating High-Resolution Maps of Leaf Water Isotopes Using IM-CRDS and IRMS Techniques. American Geophysical Union Fall Meeting.

2013

80. Scanlon, T., **Caylor, K.**, Saha, M., D'Odorico, P. (2013) Vegetation Dynamics in Response to Interannual Rainfall Variability in Southern Africa. EGU General Assembly Conference.
79. Good, S., O'Connor, M., Soderberg, K., Wang, L., **Caylor, K.** (2013) Analysis of the distribution of the isotopic composition of evapotranspiration flux in a semi-arid savanna.. EGU General Assembly Conference.
78. McCord, P.F., Evans, T.P., Dell'Angelo, J., Gower, D., McBride, L., **Caylor, K.K.** (2013) Water governance within Kenya's Upper Ewaso Ng'iro Basin: Assessing the performance of water projects. American Geophysical Union Fall Meeting.
77. Manfreda, S., Pizzolla, T., **Caylor, K.K.** (2013) Soil Water Dynamics in a Clumped Vegetation Pattern. American Geophysical Union Fall Meeting.
76. Debats, S.R., Fuchs, T.J., Thompson, D.R., Estes, L.D., Evans, T.P., **Caylor, K.K.** (2013) A Machine Learning Approach to Mapping Agricultural Fields Across Sub-Saharan Africa. AGU Fall Meeting.
75. Grossman, R.H., Guan, K., Estes, L.D., **Caylor, K.K.**, Dandois, J.P. (2013) Closing Scale Gaps in Ecological Research using Unmanned Aerial Vehicles: a meta-data synthesis. American Geophysical Union Fall Meeting.
74. Li, S., Levin, N., Soderberg, K., Dennis, K.J., **Caylor, K.K.** (2013) The triple oxygen isotope composition of leaf waters in Mpala, central Kenya. American Geophysical Union Fall Meeting.
73. Gerlein, C., Wolf, A., **Caylor, K.K.** (2013) Stable isotopes in plant physiology: using water isotopes to study water fluxes in a temperate forest. American Geophysical Union Fall Meeting.
72. Soderberg, K., Gerlein, C., Kemeny, P.C., **Caylor, K.K.** (2013) Isotopic equilibrium between precipitation and water vapor: evidence from continental rains in central Kenya. American Geophysical Union Fall Meeting.
71. Estes, L.D., Chaney, N., Herrera-Estrada, J., **Caylor, K.K.**, Sheffield, J., Wood, E.F. (2013) Spatial trends in evapotranspiration components over Africa between 1979 and 2012 and their relative influence on crop water use. American Geophysical Union Fall Meeting.
70. Evans, T.P., McCord, P.F., McBride, L., Gower, D., **Caylor, K.K.** (2013) Water Governance and Adaptation to Disturbances in Irrigated Semi-Arid Agricultural Systems. American Geophysical Union Fall Meeting.
69. Guan, K., Good, S.P., **Caylor, K.K.** (2013) Seasonal rainfall variability and its significance to terrestrial ecosystems in Africa. American Geophysical Union Fall Meeting.
68. Sheffield, J., Wood, E.F., Chaney, N., Fisher, C.K., **Caylor, K.K.** (2013) A Seamless Framework for Global Water Cycle Monitoring and Prediction. American Geophysical Union Fall Meeting.
67. O'Donnell, F.C., **Caylor, K.K.**, Gerlein, C., Bhattachan, A. (2013) Ground penetrating radar measurements show a spatial relationship between coarse root biomass and soil carbon abundance. American Geophysical Union Fall Meeting.

66. O'Donnell, F.C., **Caylor, K.K.**, Bhattachan, A., Dintwe, K., D'Odorico, P., Okin, G.S. (2013) Species-level diversity of belowground structure in savanna woody plants. American Geophysical Union Fall Meeting.

2012

65. **Caylor, K.**, Rodriguez-Iturbe, I. (2012) Scale-dependent controls on the metabolic organization of river basins. European Geophysical Union General Assembly Conference.
64. Soderberg, K., Good, S.P., O'Connor, M., King, E.G., **Caylor, K.K.** (2012) Evapotranspiration partitioning in a semi-arid African savanna using stable isotopes of water vapor. European Geophysical Union General Assembly Conference.
63. O'Donnell, F.C., **Caylor, K.K.**, D'Odorico, P., Okin, G.S., Bhattachan, A., Dintwe, K. (2012) Inter-annual rainfall variability supports coexistence of savanna tree and shrub species with dimorphic rooting strategies. 97th Ecological Society of Annual Meeting.
62. **Caylor, K.K.**, Good, S.P., Guan, K. (2012) When change is more of the same: The increasing importance of hydrological variability in drylands. American Geophysical Union Fall Meeting.
61. Guan, K., Wood, E.F., Medvigy, D., **Caylor, K.K.** (2012) Quantify the impacts of seasonal variability in precipitation on vegetation structure in Africa. American Geophysical Union Fall Meeting.
60. **Caylor, K.K.**, Estes, L.D., Evans, T., Sweeney, S. (2012) Examining Zambian Farmers' Response to Drought: A Remotely-sensed Analysis of Changes in Sowing Dates. American Geophysical Union Fall Meeting.
59. Debats, S.R., **Caylor, K.K.**, Estes, L.D., Chaney, N., Sheffield, J. (2012) Interacting Effects of Heat Stress and Soil Moisture Stress on Crop Yield Losses in Dryland Agriculture. American Geophysical Union Fall Meeting.
58. Nadeau, P., **Caylor, K.K.**, Good, S.P., Guan, K. (2012) Climate and ecosystem structure as drivers of regional fuel load accumulation and subsequent fire regimes in Africa. American Geophysical Union Fall Meeting.
57. Manfreda, S., Pizzolla, T., **Caylor, K.K.** (2012) Effects Climate Change on Water Resources Availability and Vegetation Patterns. American Geophysical Union Fall Meeting.
56. Riihimäki, C.A., Sealfon, C.D., Paine, E.N., O'Donnell, F.C., **Caylor, K.K.**, Wilcove, D.S. (2012) Princeton Science and Engineering Education Initiative: Revising Undergraduate Environmental Science Courses. American Geophysical Union Fall Meeting.
55. Estes, L.D., **Caylor, K.K.**, Choi, J. (2012) Turkers in Africa: A Crowdsourcing Approach to Improving Agricultural Landcover Maps. American Geophysical Union Fall Meeting.
54. O'Connor, M., Good, S.P., **Caylor, K.K.** (2012) A multi-year record of flux measurements in East Africa: coupling eddy-covariance with high-frequency water isotope observations. American Geophysical Union Fall Meeting.
53. D'Odorico, P., Bhattachan, A., Tatlhago, M., Dintwe, K., O'Donnell, F.C., **Caylor, K.K.**, Okin, G.S., Perrot, D., Ringrose, S. (2012) Hydrologic controls on patterns of below ground woody biomass in savannas. American Geophysical Union Fall Meeting.
52. Good, S.P., Rodriguez-Iturbe, I., **Caylor, K.K.** (2012) Stochastic modeling of 3D variability ecosystem structure and function. American Geophysical Union Fall Meeting.
51. Soderberg, K., Gitonga, J.M., **Caylor, K.K.** (2012) Establishing a water isotope framework for investigating ecosystem functioning and agricultural water use in central Kenya. American Geophysical Union Fall Meeting.

50. Evans, T., Cox, M., McCord, P., **Caylor, K.K.**, Washington-Ottombre, C., Soderberg, K., Sadri, S. (2012) Water governance, agricultural development and community-level resilience to climate change. American Geophysical Union Fall Meeting.
49. Wolf, A., Falusi, J., **Caylor, K.K.**, Sheffield, J., Wood, E.F. (2012) A GSM-based surface meteorology network in service of improved African hydrological data assimilation and drought forecasting. American Geophysical Union Fall Meeting.

2011

48. Manoli, G., Chavarro-Rincon, D., Marani, M., Putti, M., Wolski, P., Murray-Hudson, M., **Caylor, K.K.** (2011) Groundwater-vegetation interactions in the small-scale dynamics of the Okavango Delta. American Geophysical Union Fall Meeting.
47. **Caylor, K.K.**, Good, S.P., Soderberg, K., King, E. (2011) Savanna Grassland Transpiration Fluxes After A Water Pulse Using Stable Isotope And Eddy Covariance Techniques. American Geophysical Union Fall Meeting.
46. Dintwe, K., Okin, G., D'Odorico, P., **Caylor, K.K.**, O'Donnell, F.C., Bhattachan, A. (2011) Soil organic carbon and nitrogen content in savannas: Potential impacts of climate change. American Geophysical Union Fall Meeting.
45. O'Donnell, F.C., **Caylor, K.K.** (2011) Influence of rainfall climatology and coarse root structure on water use and carbon uptake by savanna trees. American Geophysical Union Fall Meeting.
44. Good, S.P., Soderberg, K., Wang, L., **Caylor, K.K.** (2011) Uncertainty in estimates of the isotopic composition of water vapor fluxes: direct comparison of techniques using laser-based analyzers. American Geophysical Union Fall Meeting.
43. Sheffield, J., Herrera-Estrada, J.E., **Caylor, K.K.**, Wood, E.F. (2011) Drought, Climate Change and Potential Agricultural Productivity. American Geophysical Union Fall Meeting.

2010

42. O'Donnell, F., **Caylor, K.** (2010) Accounting for" hot spots" and" hot moments" in soil carbon models for water-limited ecosystems. European Geophysical Union General Assembly Conference.
41. Mladenov, N., Okin, G.S., O'Donnell, F.C., D'Odorico, P., Meyer, T., Dintwe, K., **Caylor, K.K.**, Kim, S., Ringrose, S. (2010) Linking soil moisture with chemical quality of soil organic matter to evaluate belowground carbon storage in savannas. American Geophysical Union Fall Meeting.
40. Franz, T.E., Nolan, J.T., **Caylor, K.K.**, Nordbotten, J.M., Slater, L.D. (2010) Measurement and Modeling of Lateral Subsurface Transient Soil Moisture Dynamics Using Multi-Point Direct-Current Resistivity in Homogeneous Sand. American Geophysical Union Fall Meeting.
39. Good, S.P., Wang, L., **Caylor, K.K.** (2010) Isotope Techniques For The Partitioning of Evapotranspiration Into Its Constituent Components. American Geophysical Union Fall Meeting.
38. **Caylor, K.K.**, Franz, T.E., King, E., Robinson, D. (2010) Drivers of emergent vegetation pattern formation at hillslope scales in a central Kenya dryland. American Geophysical Union Fall Meeting.
37. Guan, K., Wood, E.F., **Caylor, K.K.** (2010) Multi-sensor synthesis of vegetation pattern over a large climatic gradient transect in Africa. American Geophysical Union Fall Meeting.
36. O'Donnell, F.C., **Caylor, K.K.**, D'Odorico, P., Okin, G.S., Bhattachan, A., Dintwe, K. (2010) Coarse root structure in water-limited ecosystems: Results of large-scale tree and shrub excavations across a rainfall gradient in Southern Africa. American Geophysical Union Fall Meeting.

35. Wang, L., Good, S.P., **Caylor, K.K.** (2010) A New Method to Quantify the Isotopic Signature of Leaf Transpiration: Implications for Landscape-Scale Evapotranspiration Partitioning Studies. American Geophysical Union Fall Meeting.

2009

34. Wang, L., **Caylor, K.**, Dragoni, D. (2009) Continuous measurements of water vapor isotopic compositions using an integrated cavity output spectrometer: calibrations and applications. European Geophysical Union General Assembly Conference.
33. Wang, L., **Caylor, K.**, Macko, S., Shugart, H. (2009) Integrating hydrology and biogeochemistry in drylands: the Kalahari Transect as a model ecosystem. European Geophysical Union General Assembly Conference.
32. O'Donnell, F.C., **Caylor, K.K.** (2009) Soil CO₂ Fluxes Following Wetting Events: Field Observations and Modeling. American Geophysical Union Fall Meeting.
31. Guan, K., Wood, E.F., **Caylor, K.K.**, Li, H. (2009) Attributions of Vegetation Productivity along a Climate Gradient Transect in Eastern African Savanna. American Geophysical Union Fall Meeting.
30. Franz, T.E., Nolan, J.T., **Caylor, K.K.**, Slater, L.D. (2009) Measurement and modeling of mass-infiltration events into macropores. American Geophysical Union Fall Meeting.
29. **Caylor, K.K.** (2009) Land degradation in dryland regions: An ecohydrological framework. American Geophysical Union Fall Meeting.
28. Wang, L., **Caylor, K.K.**, Good, S.P., Villegas, J.C., Breshears, D.D. (2009) Landscape-scale observations of plant water use using continuous stable isotope monitoring. American Geophysical Union Fall Meeting.

2008

27. **Caylor, K.K.**, Franz, T. (2008) Using Geophysical Measurements to Infer Subsurface Vegetation Patterns and Processes.. The 2008 Joint Annual Meeting.
26. **Caylor, K.K.**, Scanlon, T.M., Rodriguez-Iturbe, I. (2008) Ecohydrological optimization of pattern and processes in water-limited ecosystems. Water and the Environment: Proceedings of the Workshop in the Vatican Academy of Sciences.
25. Dragoni, D., **Caylor, K.K.** (2008) Decoupling Structural and Environmental Determinants of Sap Velocity Using Heat Pulse Methods. VII International Workshop on Sap Flow 846.
24. **Caylor, K.K.**, Scanlon, T. (2008) OOS 13-6: Transitions between facilitative/competitive effects of tree canopies on herbaceous productivity via impacts on energy/water balance. The 93rd Ecological Society of America Annual Meeting.
23. Franz, T., King, E., Lester, A., **Caylor, K.K.**, Celia, M., Rodriguez-Iturbe, I., Rubenstein, D.I. (2008) OOS 2-1: Differences in small-scale water balance by representative hydrologic units of central Kenya. The 93rd Ecological Society of American Annual Meeting.
22. Vinukollu, R.K., **Caylor, K.**, Wood, E.F. (2008) Evaluation of a global evapotranspiration product using FLUXNET data. American Geophysical Union Fall Meeting.
21. Franz, T.E., King, E.G., Lester, A., **Caylor, K.K.**, Nordbotten, J., Celia, M.A., Rodriguez-Iturbe, I. (2008) Ecohydrologic Modeling of Hillslope Scale Processes in Dryland Ecosystems. American Geophysical Union Fall Meeting.
20. **Caylor, K.K.**, Franz, T. (2008) Impacts of climate variability on functional diversity and species distribution in dryland ecosystems. American Geophysical Union Fall Meeting.

19. Wang, L., **Caylor, K.**, Dragoni, D. (2008) Measuring hourly ^{18}O and ^2H fluxes in a mixed hardwood forest using an integrated cavity output spectrometer. American Geophysical Union Fall Meeting.

2007

18. **Caylor, K.K.**, D'Odorico, P., Okin, G.S. (2007) Evidence and implications of soil moisture-vegetation feedbacks in semiarid savannas. American Geophysical Union Fall Meeting.
17. Lindner, G.A., **Caylor, K.K.** (2007) Assessing the Impact of Spatial Scaling on Empirical Runoff Ratio Models within a Heterogeneous Suburbanizing Watershed in Central Indiana. American Geophysical Union Fall Meeting.
16. **Caylor, K.K.**, Dragoni, D. (2007) Decoupling structural and environmental determinants of sap velocity. American Geophysical Union Fall Meeting.

2006

15. Scanlon, T.M., D'Odorico, P., Okin, G.S., **Caylor, K.K.** (2006) Effects of mean annual rainfall on landscape-level distributions of soil moisture and soil development in a semi-arid ecosystem. American Geophysical Union Fall Meeting.
14. Dragoni, D., **Caylor, K.K.**, Schmid, H. (2006) Characterization of transpiration in a deciduous forest of the US Midwest. American Geophysical Union Fall Meeting.
13. Ries, L.P., Shugart, H.H., **Caylor, K.K.**, Okin, G.S., Kgope, B. (2006) Nutrient and light limitations on grass productivity in a southern African savanna. American Geophysical Union Fall Meeting.

2005

12. **Caylor, K.K.**, Rodriguez-Iturbe, I. (2005) Spatial variation of surface hydrological dynamics in semi-arid landscapes. American Geophysical Union Spring Meeting.

2004

11. Scanlon, T.M., **Caylor, K.K.**, Rodriguez-Iturbe, I. (2004) Ecohydrological significance of observed power-law distributions of tree canopy cluster sizes for savanna vegetation. American Geophysical Union Fall Meeting.
10. **Caylor, K.K.**, Rodriguez-Iturbe, I. (2004) Coupling ecohydrological patterns and processes in semi-arid landscapes. Proceedings of the British Hydrological Society's International Conference on Hydrology: Science and Practice for the 21st century.
9. **Caylor, K.**, D'Odorico, P., Rodriguez-Iturbe, I. (2004) Horizontal and vertical variability of soil moisture in savanna ecosystems. American Geophysical Union Fall Meeting.

2003

8. Hély, C., **Caylor, K.K.**, Alleaume, S., Swap, R.J., Shugart, H.H. (2003) Estimating emissions of carbonaceous compounds from southern african fires: a re-analysis of modeling techniques. EGS-AGU-EGU Joint Assembly.
7. Manfreda, S., **Caylor, K.**, Rodriguez-Iturbe, I. (2003) On The Soil Moisture Dynamics: Does Simplification Alter Soil Moisture Prediction?. American Geophysical Union Fall Meeting.

6. Shugart, H.H., **Caylor, K.K.**, Scanlon, T.M., Rodriguez-Iturbe, I. (2003) Assessing the role of spatial pattern in governing ecohydrological interactions and vegetation dynamics in semi-arid savannas.. American Geophysical Union Fall Meeting.

2002

5. **Caylor, K.K.**, Shugart, H.H. (2002) Simulated water-use of heterogeneous vegetation patches along the Kalahari transect. American Geophysical Union Fall Meeting.

2001

4. **Caylor, K.K.**, Shugart, H.H. (2001) Variation in vegetation structure along a large moisture gradient: The role of rainfall variability in maintaining spatial heterogeneity. American Geophysical Union Fall Meeting.
3. Shugart, H.H., **Caylor, K.K.**, Dowty, P.R. (2001) Ecological Scale Issues in Land/Atmosphere Coupling-Examples from SAFARI 2000. American Geophysical Union Fall Meeting.
2. Scanlon, T.M., Albertson, J.D., **Caylor, K.K.**, Williams, C.A. (2001) Assessing Fractional Tree, Grass, and Bare Soil Cover from NDVI and Rainfall Time Series along the Kalahari Transect, Africa. American Geophysical Union Fall Meeting.
1. **Caylor, K.K.**, Dowty, P.R., Shugart, H.H. (2001) Relationship between small-scale structural heterogeneity and simulated vegetation productivity across a regional moisture gradient. American Geophysical Union Fall Meeting.